

Does Momentum Survive Transaction Costs in Korea?

Survivorship-Free Evidence Net of a Time-Varying Securities Transaction Tax

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This version: August 2026

Abstract

I test whether cross-sectional price momentum is profitable in the Korean equity market once realistic frictions and standard biases are removed. Using a survivorship-free universe of 3,361 securities from Compustat Global Security Daily over 2010–2026 — including every firm that delisted during the sample — I evaluate twelve long-only momentum specifications and the value-weighted winner-minus-loser (WML) spread against a true buy-and-hold benchmark, net of Korea’s statutory securities transaction tax (STT), commissions, and slippage. The WML spread is 6.0% per year gross but statistically indistinguishable from zero ($t = 0.60$), and it is insignificant in every sub-period examined. No long-only specification beats buy-and-hold net of costs; the most concentrated variants lose up to 19% per year. None survives White’s (2000) Reality Check for data-snooping ($p = 0.67$). A cost decomposition shows the shortfall is driven primarily by the weakness of the underlying signal rather than by the transaction tax: even with all costs set to zero, momentum underperforms buy-and-hold, while the STT contributes roughly one percentage point per year. The results reinforce and modernize earlier findings of weak or negative momentum in Korea (Chui, Titman, and Wei, 2000; Chae and Eom, 2009), using survivorship-free data, a time-varying statutory cost schedule, and multiple-testing corrections.

Keywords: *momentum, market anomalies, transaction costs, survivorship bias, Korean equity market, data snooping*

JEL classification: *G11, G12, G14*

1. Introduction

Cross-sectional price momentum — the tendency of recent winners to keep outperforming recent losers — is among the most durable and widely replicated anomalies in empirical asset pricing. Since Jegadeesh and Titman (1993) first documented that buying past winners and selling past losers earns significant returns over three-to-twelve-month horizons, momentum profits have been found across most developed and emerging markets (Rouwenhorst, 1998; Griffin, Ji, and Martin, 2003; Asness, Moskowitz, and Pedersen, 2013). Yet the anomaly is not universal. Chui, Titman, and Wei (2000) find that momentum obtains across Asian markets with the notable exceptions of Japan and Korea, and Chae and Eom (2009) document outright negative momentum profits in Korea. These exceptions raise a question that this paper addresses directly: is momentum genuinely absent in Korea, or does its apparent weakness reflect measurement choices — survivorship bias, the treatment of transaction costs, and multiple testing — that the earlier literature did not fully control?

I revisit the Korean momentum question with three methodological commitments that jointly distinguish this study from prior work. First, the universe is survivorship-free: it is drawn point-in-time from Compustat Global Security Daily and retains every firm that delisted during the sample, so the strategy is never advantaged by hindsight knowledge of which firms survived. Second, costs are modeled with Korea’s actual statutory securities transaction tax (STT), which changed seven times over the sample period, from 0.30% down to 0.15% and back to 0.20% in 2026, alongside commissions and slippage. Third, statistical inference is corrected for the number of specifications examined, using the Deflated Sharpe Ratio (Bailey and López de Prado, 2014) and White’s (2000) Reality Check, so that an apparently strong result cannot arise merely from searching across many strategy variants.

The central findings are as follows. The value-weighted winner-minus-loser spread is 6.0% per year gross but statistically indistinguishable from zero ($t = 0.60$), and it is insignificant in every sub-period examined, including pre- and post-2019 splits. Across twelve long-only specifications spanning concentration (top 10, 20, 50 names), weighting (equal and value), and formation lag (skipping the most recent month or not), none beats a true buy-and-hold of the liquid universe net of costs; losses scale monotonically with concentration in extreme winners, reaching -19% per year for the most aggressive variant. No specification survives White’s Reality Check ($p = 0.67$). Finally, a cost decomposition isolates the source of the shortfall: with all costs set to zero, momentum still underperforms buy-and-hold, indicating that the weakness of the underlying signal — not the transaction tax — is the primary driver. The STT contributes on the order of one percentage point per year, deepening but not creating the underperformance.

These results contribute to two literatures. They reinforce and modernize the finding of weak-to-negative Korean momentum (Chui, Titman, and Wei, 2000; Chae and Eom, 2009) with survivorship-free data and data-snooping-robust inference, showing the earlier conclusion is not an artifact of a particular sample or specification. And they speak to the transaction-cost literature on anomaly implementability (Novy-Marx and Velikov, 2016): by decomposing the cost wedge and using Korea’s uniquely sharp sell-side STT, the paper shows that in this market costs are a secondary rather than a primary explanation for momentum’s failure.

2. Data and Sample Construction

Sample. The sample consists of daily security-level data for Korean common stocks from Compustat Global Security Daily over January 2010 through August 2026. The raw extract contains 3,556 securities; after requiring a minimum trading history and valid pricing fields, 3,361 securities enter the analysis. Crucially, the sample retains all firms that delisted during the period. Each security enters the panel at its first valid trading date and exits at its last, so the cross-section on any given date contains only firms actually trading that day — the panel is genuinely point-in-time and survivorship-free.

Prices and adjustments. Prices are adjusted for splits and dividends using the daily adjustment factor and total-return factor, yielding a total-return price series. The liquid universe on each rebalance date comprises the 150 securities with the highest trailing 60-day median traded value, computed using only information available on that date. This screen restricts attention to economically tradeable names while remaining strictly point-in-time.

Transaction costs. Korea levies a securities transaction tax on the sell side only, which makes it an unusually sharp setting for cost analysis because the tax falls precisely on the high sell-turnover that momentum requires. The statutory KOSPI rate (inclusive of the rural development special tax) changed repeatedly over the sample; the schedule applied in the simulation is 0.30% through mid-2019, then 0.25%, 0.23%, 0.20% (2023), 0.18% (2024), 0.15% (2025), and 0.20% from January 2026. Each simulated sale pays the rate in force on the trade date. Commissions and slippage are added on both sides, and a square-root market-impact term scaled to each name’s liquidity is available for robustness.

3. Methodology

3.1 Momentum specifications and benchmark

Momentum is formed on the trailing twelve-month total return, optionally skipping the most recent month to avoid short-term reversal (Jegadeesh, 1990; Lehmann, 1990). On each monthly rebalance date, eligible names are the intersection of the liquid universe and those alive that day; the strategy holds the top N by momentum, equal- or value-weighted. I evaluate twelve specifications spanning $N \in \{10, 20, 50\}$, equal versus value weighting, and formation lag of zero or one month. The benchmark is a true buy-and-hold: the liquid universe is selected once at the start of the evaluation period and held thereafter, with positions exited only upon delisting. This is the economically correct comparison because a periodically rebalanced equal-weight benchmark would itself churn the sell-side tax and understate the passive alternative.

3.2 Simulation

Backtests run on a purpose-built event-driven engine designed for survivorship-free panels. Signals are computed on each day’s close and orders execute at the next day’s open, precluding look-ahead. Positions respect Korean tick sizes and daily price limits. A stock that delists is force-liquidated at its last traded price, with a -30% Shumway (1997) haircut available as a robustness check. The engine tracks cash and share positions exactly and never holds a position in a non-trading name, eliminating the phantom-holding bias that corrupts naive survivorship-free implementations.

3.3 Statistical inference

Because a strategy that is the best of many trials can appear profitable by chance, I apply three layers of inference. Pairwise outperformance over buy-and-hold is tested with the Jobson–Korkie (1981) statistic with Memmel’s (2003) correction. The Deflated Sharpe Ratio (Bailey and López de Prado, 2014) adjusts each specification’s Sharpe ratio for the number of trials, sample length, skewness, and kurtosis. Finally, White’s (2000) Reality Check, implemented via a stationary block bootstrap, tests the joint null that the best specification does not outperform the benchmark once the entire search is accounted for. Factor-adjusted alphas are estimated against market and size factors constructed from the same survivorship-free universe, with Newey–West (1987) standard errors.

4. Results

4.1 The momentum premium

Table 1 reports annualized returns of value-weighted momentum deciles, D1 (losers) through D10 (winners). The pattern is not monotonic: returns do not rise cleanly from losers to winners, and the extreme-winner decile (D10) underperforms the intermediate deciles D8 and D9 — the signature of short-term reversal among stocks with the most extreme prior gains. The winner-minus-loser (D10–D1) spread is 6.0% per year gross of costs, with a t-statistic of 0.60: statistically indistinguishable from zero even before any trading friction. Splitting the sample does not rescue the premium; the WML spread is insignificant in the first half ($t = 0.25$), the second half ($t = 0.60$), and in pre- and post-2019 sub-periods ($t = 0.07$ and 0.79 , respectively). The premium is not a once-strong effect that decayed — it is never statistically distinguishable from zero.

Table 1. Value-weighted momentum decile returns (annualized, gross).

Decile	Ann. return
D1 (losers)	1.6%
D2	7.3%
D3	7.6%
D4	3.4%
D5	9.4%
D6	9.8%
D7	6.5%
D8	12.8%
D9	14.5%
D10 (winners)	7.9%
WML (D10–D1)	6.0% ($t = 0.60$)

4.2 Long-only specification sweep

Table 2 reports the twelve long-only specifications against true buy-and-hold, which earned 2.1% per year with a –31% maximum drawdown — a modest figure consistent with the well-documented “Korea discount” over this period. No specification beats the benchmark on CAGR. Losses scale monotonically with concentration in extreme winners: the most diversified variant (top 50, value-weighted) is roughly flat, while the most aggressive (top 10, equal-weighted, no skip) loses 19.0% per year with a –99% drawdown. Three specifications underperform the benchmark with statistical significance; none outperforms significantly, and every Deflated Sharpe Ratio is near zero.

Table 2. Long-only momentum specifications, net of time-varying tax.

Specification	CAGR	Sharpe	Max DD	DSR
Buy & hold (benchmark)	2.1%	-0.10	-31.2%	—
top50·value·skip0	-0.3%	-0.12	-81.5%	0.13
top50·value·skip21	-1.8%	-0.17	-83.2%	0.09
top50·equal·skip21	-5.5%	-0.34	-80.6%	0.02
top20·value·skip0	-9.6%	-0.35	-96.4%	0.02
top10·value·skip0	-17.1%	-0.47	-98.7%	0.01
top10·equal·skip0	-19.0%	-0.59	-98.8%	0.00

Note: representative rows shown; full twelve-specification table in the online appendix. DSR is the Deflated Sharpe Ratio probability.

White's Reality Check across the full search yields a p-value of 0.67: once the number of specifications tried is accounted for, no strategy beats buy-and-hold beyond what search luck would produce.

4.3 Is the tax the problem, or the signal?

A natural hypothesis is that Korea's sell-side tax is the binding constraint: momentum churns its portfolio, and the tax falls on every sale. Table 3 tests this directly by re-running the strategy with costs progressively removed. The result is unambiguous. Even frictionless — with the tax, commissions, and slippage all set to zero — the best specification returns only 1.4% per year, still below the 2.1% earned by buy-and-hold, and the most aggressive specification still loses 17.6% per year. The transaction tax accounts for roughly one percentage point of annual drag. The primary driver of momentum's failure in Korea is therefore the weakness of the underlying signal, not the cost of trading it; costs deepen a shortfall they do not create.

Table 3. Cost decomposition (annualized CAGR).

Specification	Frictionless	No tax	Full costs	Tax drag
Best (top50·value·skip0)	1.4%	0.6%	-0.3%	0.9%/yr
Most aggressive (top10·equal)	-17.6%	-18.0%	-19.0%	1.0%/yr
Buy & hold reference	—	—	2.1%	—

4.4 Robustness

The sign of the result is invariant to cost assumptions. Varying slippage from 0 to 20 basis points and applying a -30% delisting haircut leaves every long-only specification below the benchmark. Factor-adjusted alphas confirm the picture: the best long-only specification has an annualized alpha of -1.5% ($t = -0.35$) against market and size factors, and the WML spread's alpha of 5.9% carries a t-statistic of only 0.94 — economically visible but statistically unreliable, consistent with the raw spread.

5. Discussion and Limitations

The evidence supports a specific and bounded conclusion: in the Korean equity market over 2010–2026, cross-sectional momentum offers no statistically reliable premium, gross or net, and no long-only implementation is competitive with passive buy-and-hold after realistic costs. This extends Chui, Titman, and Wei (2000) and Chae and Eom (2009) with survivorship-free data and data-snooping-robust inference, and it locates the cause: signal weakness compounded by, but not caused by, the sell-side transaction tax.

Several limitations bound the interpretation. First, the analysis is long-only; the academic momentum factor is long–short, and short-sale constraints in Korea make the long–short version largely

uninvestable for most participants, so this study answers the practically relevant rather than the purely theoretical question. Second, an attempt to exploit the staggered STT schedule as a natural experiment for the cost mechanism proved infeasible: the tax changes coincide with major market-regime shifts — most severely the 2025–2026 artificial-intelligence and semiconductor rally, during which Samsung Electronics and SK Hynix rose from 34% to over 50% of KOSPI market capitalization — so statutory-rate variation cannot be cleanly separated from contemporaneous conditions. This is reported as a limitation rather than a result. Third, the value factor is omitted from the factor model for want of a fundamentals extract; incorporating book-to-market is left to future work.

6. Conclusion

Using survivorship-free data, a time-varying statutory cost schedule, and multiple-testing corrections, this paper shows that cross-sectional momentum is not a reliable source of return in the Korean equity market. The winner-minus-loser spread is indistinguishable from zero even before costs; no long-only specification beats buy-and-hold net of costs; and none survives a data-snooping correction. A cost decomposition attributes the failure primarily to signal weakness rather than to Korea's sell-side transaction tax, which contributes on the order of one percentage point per year. The results discipline a widely assumed anomaly in a specific market and illustrate the value of honest, survivorship-free, cost-aware, search-corrected evaluation in distinguishing genuine premia from artifacts.

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